Bags, Vectors & Transformers

Fine-Tuning an LLM · Bonus, Part 18

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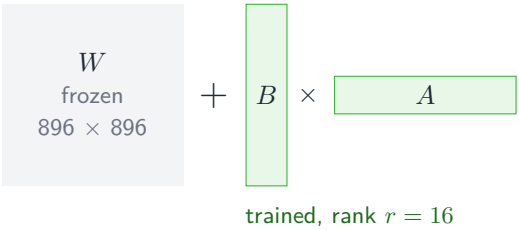
Prompting Changes the Question, Fine-Tuning Changes the Model

	What changes	Labels	Good for
Prompting (Day 4)	the prompt	a test set	trying things, small jobs
Fine-tuning BERT (Day 3)	a small encoder	hundreds	classification at scale
Distillation (Day 4)	an encoder, on LLM labels	a test set	scale with few human labels
Fine-tuning an LLM	a small add-on to the LLM	hundreds	text as output, fixed formats

Worth it when the output is text in a fixed form, or a long codebook prompt would travel with every text.

Not for new facts. It changes how a model behaves more than what it knows.

LoRA: Train a Small Add-On, Not the Whole Model



- W stays frozen; only two thin matrices are trained (Hu et al., 2022)
- for Qwen2.5 0.5B: **under 1%** of the parameters
- the result is an **adapter** of a few megabytes, switchable on and off
- **QLoRA** stores the frozen model in 4 bits: 7B models fit on a Colab GPU (Dettmers et al., 2023)

Training Data: Conversations, Graded on the Answer Only

```
<|im_start|>system
You code US bill titles by policy area.
Reply with the category name only.<|im_end|>
<|im_start|>user
Bill title:  To amend the Internal Revenue
Code to ...<|im_end|>
<|im_start|>assistant
Taxation<|im_end|>
```

- 200 coded titles per area, the same data as Days 3 and 4
- a **short** instruction: the codebook moves into the weights
- gray tokens get the label −100: **not in the loss**
- the model learns the answer, and when to stop

What to Compare

The same 100 test titles

- Qwen2.5 0.5B with the codebook prompt
- the same model, fine-tuned, short prompt
- the same model with 8 examples in the prompt
- your Day 4 strategies (1.5B) and Day 3 DistilBERT

Accuracy, invalid answers, prompt length. The fine-tuned model needs no codebook in the prompt, so every text costs fewer tokens.

Fine-tuned smaller models regularly beat much larger models used zero-shot on text classification (Bucher & Martini, 2024).

How much data? Train on 100, 400 and 800 titles: the curve shows whether coding more is worth it.

What Else Changed?

A model fine-tuned on one narrow task becomes a **specialist**: asked something else, it may answer with a policy area anyway.

Because the adapter is separate, you can switch it off and compare:

```
with ft_model.disable_adapter():
print(answer(ft_model, question))
```

For a research instrument that's fine, as long as you use it only for what it was trained on. LoRA tends to forget less than full fine-tuning (Biderman et al., 2024).

New languages, new eras, new text types: validate before you trust it there.

Saving, Sharing, Reporting

Save
Only the adapter: a few MB. Reload it on top of the same base model, or merge it into the weights.

Share
Push it to the Hub, unless it was trained on confidential or personal texts: models can leak their training data.

License
The adapter inherits the base model's license. Qwen2.5 0.5B is Apache 2.0; others set conditions.

Report: base model and version, training data and its size, LoRA rank, alpha and target layers, epochs, learning rate, seed, and test scores. For larger jobs, TRL and Unsloth wrap the same steps.

In the Notebook

- When fine-tuning an LLM is worth it, and when BERT or distillation is enough
- Coded titles as chat conversations, with the prompt masked out of the loss
- LoRA fine-tuning of Qwen2.5 0.5B in a few minutes on a T4
- Prompted vs. few-shot vs. fine-tuned; how much training data you need
- Switching the adapter off: what else changed?
- Saving, reloading and sizing the adapter

`bonus/notebooks/18_fine_tuning_an_llm_exercises.ipynb` · 5 exercises · GPU